



ME 300: Engineering Thermodynamics

Example: Compressors and Turbines

Compressor

An air compressor running with a constant mass flow rate has the following inlet and exit conditions. Calculate the power required to drive the compressor.

$$P_1 = 100 \text{ kPa}$$

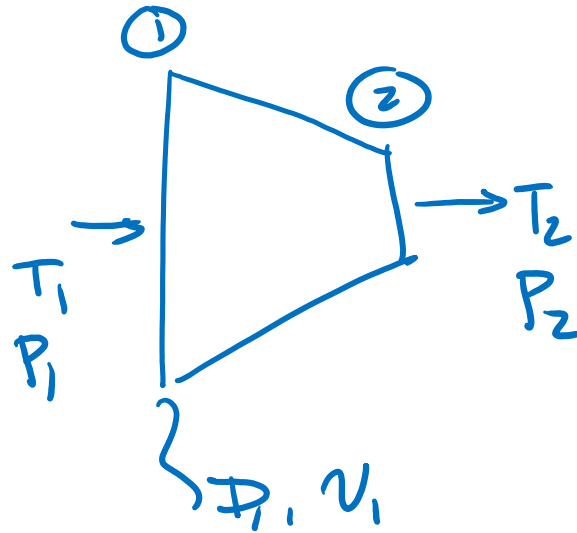
$$T_1 = 300 \text{ K}$$

$$\text{Inlet diameter} = 0.3 \text{ m}$$

$$\text{Inlet velocity} = 12 \text{ m/s}$$

$$P_2 = 350 \text{ kPa}$$

$$T_2 = 435 \text{ K}$$



Assume

- steady-flow
- air = ideal gas, $c_p = \text{const.}$
- quasi-st.
- $\Delta \text{ke} = \Delta \text{pe} = 0$, $\dot{Q} = 0$

$$\Rightarrow \dot{m} \Delta h = -\dot{W}$$



Which of the following assumptions is **NOT** appropriate for this problem?

- A. Quasi-equilibrium ✓
- B. $\Delta KE = \Delta PE = 0$ ✓
- C. Ideal gas ✓
- D. They are all appropriate

Compressor

$$\rightarrow \Delta h = c_p \Delta T \rightarrow \text{if ideal gas, } c_p = \text{const.}$$

$$\Delta h = c_p (T_2 - T_1)$$

$$= 1001 (435 - 300)$$

$$\Delta h = 136215 \frac{\text{J}}{\text{kg}}$$

$$\rightarrow \dot{m} = \rho_1 v_1 A_1 \rightarrow \rho_1 = \frac{P_1}{RT_1} = \frac{100000}{(287)(300)} = 1.153 \frac{\text{kg}}{\text{m}^3}$$

$$= 0.982 \frac{\text{kg}}{\text{s}} \quad v_1 = 12 \text{ m/s}$$

$$A_1 = \frac{\pi}{4} D_1^2 = 0.071 \text{ m}^2$$

Compressor

$$\Rightarrow \dot{m} \Delta h = -\dot{W}$$

$$\dot{W} = -\dot{m} \Delta h$$

$$= -(0.982)(136215)$$

$$\dot{W} = -133811.6 \text{ W}$$